ulta Ukraine Leadership & Technology Academy

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State of the Art in Technology: AI
Lecture Notes, Day 1

Main point of this session:
Get excited about how AI can help you learn faster and change the world.

Learning objectives for today:
- Formulate clear and effective prompts.
- Evaluate responses for accuracy and reliability.
- Get explanations in a variety of subjects.
- See pipeline of AI: data, model, train, inference.

Day 1 is all about getting excited about AI. Time is of the essence. Inspiring and concise.

[Checklist: print 20 copies of “Resources” and “The Bitter Lesson,” launch SDXL 1.0]

***The Energy in Creation*, by Laker Newhouse and GPT-2 (as students walk in)**

[Play song on repeat.]

Welcome, everybody! I like music, so in this class we start every day with a short song. I composed that song with GPT-2. That’s right: GPT-2.

Why learn AI? (25 mins, until 10:00 / 12:10)

Why does AI matter for you? How will AI become a superpower for you in your life?

Maybe you want to use AI to advance science. Maybe you want to use AI to bring a free, world class education to everyone, everywhere. Maybe you’ve seen ChatGPT all over the news. Maybe you’ve heard that AI helped computers beat humans in chess, then beat those computers, then beat those computers that beat those computers. Maybe you’ve heard about the reason we care about AI winning in chess in the first place. Maybe you want to learn about AI because you have heard that the math behind it is beautiful.

Every one of you is a finalist in ULTA. You were chosen carefully and precisely to be here because we saw in you a desire to make big changes in your community and country. We want you to be the next frontier of technology leaders. The world is facing big problems. There is a huge potential that AI can help to solve these problems. Already AI has rocketed up in capabilities faster than anyone could have predicted. AI breakthroughs are helping researchers cure diseases faster. AI breakthroughs are helping people access information, every day, that would have been buried in a library twenty years ago. And of course AI breakthroughs are helping to fight the war in Ukraine.

Maybe the next breakthrough will be you. That's why we are teaching you AI. We are not here to make the next Instagram filter. That's a solved problem. That's a problem that doesn't uplift people and communities. We are here together for four days because we want to give you a new tool to be the next leader in your community.

Let me show you some specific examples. These come from moments in my life when I said "WOW." [Write each example on whiteboard.]

1. Atari Deep Q learning (AI can play simple games) - 2013 [stop at 2:02]

Why care about games in the first place? Because big problems in the world can often be framed as games. Take designing a new molecule to be a drug that cures a disease. This is a problem that requires searching a vast space of possibilities to find the best "move." Or take physics. This is a "game" that humans have been playing for centuries. Our goal in physics is to learn how the universe works. We are trying to figure out the rules of the game, learning only from examples, exploration, and experiments. This is a problem that requires long-term planning and seeing patterns in unexpected places. What if we could build an AI that could play games at a superhuman level? What if we could teach an AI to play a game without ever telling it the rules? Then games could become a training ground for developing intelligence, for learning how to search among trillions of possible "moves," and for learning long-term planning. That's exactly our next example. (Mention: MuZero is the model that actually learned without rules; AlphaZero learned purely by self-play.)

2. AlphaZero (AI can play complex games) - 2017 [learned from scratch via self-play]
   [watch until to 11:34; hype up Stockfish vs. AlphaZero queen blunder]
3. AlphaFold (AI can do science) - 2020 [solved grand challenge in biology]
   [watch up to 1:00]

Proteins are what make everything in our bodies work. But they have a complicated 3D structure. Imagine that there's a super complicated tangle of headphone wires. Those wires are the protein. But those wires influence everything about how that protein will interact with its neighbors, and how a drug might be able to cure a disease caused by that protein. We need to know the structure. [Show video.] Already scientists are using AlphaFold to push the frontier of drug development.

But we're forgetting something. I've liked AI for many years, I've followed AI for many years, but what really got me thinking was ChatGPT.

4. [GPT-4](GPT-4) (AI can reason) - 2023

What stunned me about ChatGPT is not that it can write essays, compose poems, code algorithms, or any specific task, but its generality. Every other AI we've seen is really good at one task. Not GPT. Instead GPT is the closest humanity has come to general intelligence: one system, like our brain, that can do anything. What if GPT really were an artificial general intelligence? How would artificial general intelligence affect our society? To me, these questions feel as important as climate change, and probably more urgent. On Day 4 we will return to discuss societal implications.

Activity: why do you want to learn AI? (4 minutes, until 10:04 / 12:14)

Please take out a pen and paper. I want you to write down why you want to learn about AI. These reasons are for you. You won't need to read them aloud later. Let's take 3 minutes.

The purpose of this class is to teach you what AI is and how you can use it to do things that would have looked like magic ten years ago. We'll see the math behind AI, because AI is not magic, it is math. This class is going to be interactive. You don't learn something when I say it, but when you do it.

Activity: stable diffusion (4 minutes, until 10:08 / 12:18)

And last thing, this class is going to be a lot of fun. [Show picture of students studying AI.] Woah! You see this picture? It looks like it is from an artist. Let's have some fun. Let's workshop a prompt together and generate new pictures while we do the next activity. [Workshop prompt for one minute, type prompt in, select style "fantasy art" or "anime." Generate batch of 4.]

Activity: making ChatGPT into your personal tutor (27 minutes, until 10:35 / 12:45)

And now, let's start. I'm so excited to have the opportunity to teach you AI for four days. But, soon enough, we will part ways. I want to start by giving you the tools you need to continue learning about AI after I'm gone.

In particular, I want to emphasize one specific use of ChatGPT: to learn. Let me ask you, have you used ChatGPT before to teach you something? If so, how did it go? (3 mins)

I want to show you three specific examples. (Spend 5 minutes, 15 minutes, 5 minutes.)

1. Be my dialogue partner in a new language

(Goals: give unexpected example, expand creativity for what you can ask GPT, show that specific wording like “dialogue partner” makes prompts much more effective.)

Students pull out laptops, log on to chat.openai.com, enter in a prompt for themselves with one of the following choices.

2. Explain {calculus derivatives, linear algebra} in considerable technical depth

(Goals: GPT can teach you math up through intro college courses, phrase “in considerable technical depth” works great to pull out deeper explanations; and let students try substituting “to an advanced high school student” for a different, perhaps better result; the point is you can specify an audience for GPT to get the best response.)

Ask: What is the difference you saw when you used “in considerable technical depth” vs. “to an advanced high school student?”

Students close laptops. I share screen. Let’s see how you how GPT can teach you about AI. Let’s use an example called Word2Vec: explain “king - man + woman = queen.”

3. Can you explain Word2Vec? I'm an advanced high school student.

Address that GPT-4 is significantly better than GPT-3. But, there are other models online now that are free and just about as good as GPT-4. Show claude.ai on the same prompt.

GPT is an expert in many subjects. It can also be a great teacher. I get the feeling that the bottleneck usually isn’t GPT, but rather the questions I think to ask it.

Important note: GPT is not always correct. You should double check its responses. No, really, I double check its responses. Let’s ask GPT-4 whether 17069 = 13*13*101 is prime. (See Chen et al.)

But GPT is also the best teacher you all have for free.

That’s not to say it’s the only way I want you to learn. I’ve compiled a list of resources that I would like to give you now. I have specifically curated each item on this list to be a resource that has made a big impact in my life. I’ve left plenty of resources off the list, because I want to only give you the top ways you can continue to supercharge your learning once you return home. [Hand out 10 printed copies of “Resources”]

Check on stable diffusion (3 mins, until 10:38 / 12:48)

Let’s look at whether our stable diffusion is finished. Ah, looks like it is! So beautiful. Which of these four images do you like the most? Let’s pick your favorite.

And that will be one of the prizes at the auction: a half hour of generating images with this cutting-edge model, released this month.

[During break, print out 10 copies in color. Hand one to each student.]

Here's a tangible object that only AI could make.

— BREAK (5 minutes) — until 10:45 / 12:55

Pipeline of Machine Learning (15 minutes, until 11:00 / 1:10)

One moment for terminology. AI is a general term for any sort of intelligent computer. The breakthroughs in the last ten years are mostly due to machine learning. This class is going to focus on machine learning. Tomorrow we'll go into much more depth; today, let's look at the pipeline for any machine learning task: data, model, training, inference.

[Write on whiteboard: "data, model, training, inference"]

By the way, in AI class we'll be using our computers a lot. I want to create one ground rule. If I ask you to close your computer and put it away, please do so. If I ask you to lower your screen, it means it's fine to keep it running, but almost close it so I know you are listening.

Ask students to load Interactive Visualization of Linear Regression. Turn on "squared error." Demo on my screen. Explain what's going on when we add a point.

Explain the four steps as they apply here:

Step 1: get data points.

Step 2: choose the model (in this case, a line).

Step 3: train (fit the line).

Step 4: use the line to predict outcome for new data.

This is a basic example, but it shows you the same pipeline used by much more complicated models. ChatGPT, AlphaZero, all these models still need to get data, choose the model, train, and predict.

Let's admit it. Linear regression is cool, but not that cool. It's boring because there there is one data dimension (x), and two parameters in the model (slope, intercept). Modern neural transformers (a kind of neural network behind GPT) use data dimensions more like 10,000. They have hundreds of billions of parameters. We will have a look at the math behind that next time.

Activity: read The Bitter Lesson (20 minutes, until 11:20 / 1:30)

Minutes 90 - 110: read “The Bitter Lesson” by Rich Sutton.

One student reads aloud one section (delimited by newlines), then I ask everybody to summarize what the author is saying. Repeat.

Takeaways:

1. Size matters in AI.
2. Lots of early methods tried to bake human expertise into models. Now we know that AI should be general purpose and learn for itself. That's what we'll focus on in this class.
3. AI as a field is not set in stone. People regularly and vigorously debate it.

You are future leaders. You are the ones who will decide when AI is used, how it is used, and whether it is used to benefit only some or our whole society. More on that on Day 4. Tomorrow, we will learn the key math behind machine learning, including neural networks. See you tomorrow.